

6	(a)	(i)	(by Fleming's left hand) magnetic force is right [M1] So electric force is left and so left plate is lower potential [A1]	
6	(a)	(ii)	$Bev = eE \Rightarrow v = E/B = 1.50 \times 10^4 / 0.020$ [C1] $= 7.5 \times 10^5 \text{ m s}^{-1}$ [A1]	
6	(a)	(iii)	The other ions are <u>deflected</u> (and so do not arrive at the slit). Only ions of certain "chosen" velocity can emerge from the slit.	B1 B1
6	(b)	(i)	$Bqv = mv^2/r \Rightarrow r = mv/Bq = (5.81 \times 10^{-26})(7.5 \times 10^5) / [(0.020)(1.60 \times 10^{-19})]$ [C1] (allow ecf for v) $= 13.6 \text{ m}$ [A1]	
6	(b)	(ii)	1. no. no [A1]	